class19

Aiden Quisenberry (PID:A19083392)

```r
library(bio3d)

seq <- read.fasta("A19083392_mutant_seq.fa")
seq
```

```
               1        .         .         .         .         .         60
wt_healthy     MEHQLLCCEVETIRRAYPDANLLNDRVLRAMLKAEETCAPSVSYFKCVQKEVLPSMRKIV
mutant_tumor   MEHQLLCCEVETIRRAYPDANLLNDRVLRAMLKAEETCAPSVSYFKCVQKEVLPSMRKIV
               ************************************************************
               1        .         .         .         .         .         60

              61        .         .         .         .         .        120
wt_healthy     ATWMLEVCEEQKCEEEVFPLAMNYLDRFLSLEPVKKSRLQLLGATCMFVASKMKETIPLT
mutant_tumor   ATWMLEVCEEQKCEEEVFPLAMNYLDRFLSLEPVKKSRLQLLGATCMFVASKMKETIPLT
               ************************************************************
              61        .         .         .         .         .        120

             121        .         .         .         .         .        180
wt_healthy     AEKLCIYTDNSIRPEELLQMELLLVNKLKWNLAAMTPHDFIEHFLSKMPEAEENKQIIRK
mutant_tumor   AEKLCIYTDNSIRPEELLQMELLLVNKLKWNLAAMTPHDFIEHFLVKMPEAEENKQIERK
               ********************************************* *********** **
             121        .         .         .         .         .        180

             181        .         .         .         .         .        240
wt_healthy     HAQTFVALCATDVKFISNPPSMVAAGSVVAAVQGLNLRSPNNFLSYYRLTRFLSRVIKCD
mutant_tumor   HRQTFVALCATDVKFISNPPSMVAAGSVVAAVQGYNLRSPNNFLSYYRLTRFLSRVIKCD
               * ******************************** ************************
             181        .         .         .         .         .        240

             241        .         .         .         .         .    295
wt_healthy     PDCLRACQEQIEALLESSLRQAQQNMDPKAAEEEEEEEEEVDLACTPTDVRDVDI
```

```
mutant_tumor   PDCLRACQEQIEALLESSLRQAQQNMDPKAAEEEEEEEEEVDLACTPTDVRDVDI
               ****************************************************
             241        .         .         .         .         .    295

Call:
  read.fasta(file = "A19083392_mutant_seq.fa")

Class:
  fasta

Alignment dimensions:
  2 sequence rows; 295 position columns (295 non-gap, 0 gap)

+ attr: id, ali, call
```

```
cyclin <- get.seq("NP_444284.1")
```

```
Warning in get.seq("NP_444284.1"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.
```

```
cyclin
```

```
              1        .         .         .         .         .         60
NP_444284.1   MEHQLLCCEVETIRRAYPDANLLNDRVLRAMLKAEETCAPSVSYFKCVQKEVLPSMRKIV
              1        .         .         .         .         .         60

             61        .         .         .         .         .         120
NP_444284.1   ATWMLEVCEEQKCEEEVFPLAMNYLDRFLSLEPVKKSRLQLLGATCMFVASKMKETIPLT
             61        .         .         .         .         .         120

            121        .         .         .         .         .         180
NP_444284.1   AEKLCIYTDNSIRPEELLQMELLLVNKLKWNLAAMTPHDFIEHFLSKMPEAEENKQIIRK
            121        .         .         .         .         .         180

            181        .         .         .         .         .         240
NP_444284.1   HAQTFVALCATDVKFISNPPSMVAAGSVVAAVQGLNLRSPNNFLSYYRLTRFLSRVIKCD
            181        .         .         .         .         .         240

            241        .         .         .         .         .    295
NP_444284.1   PDCLRACQEQIEALLESSLRQAQQNMDPKAAEEEEEEEEEVDLACTPTDVRDVDI
```

```
            241        .         .         .         .         .     295

Call:
  read.fasta(file = outfile)

Class:
  fasta

Alignment dimensions:
  1 sequence rows; 295 position columns (295 non-gap, 0 gap)

+ attr: id, ali, call
```

```r
attributes(seq)
```

```
$names
[1] "id"   "ali"  "call"

$class
[1] "fasta"
```

```r
comp <- conserv(seq)
mutantpos <- which(comp != 1)
seq$ali[,mutantpos]
```

```
             [,1] [,2] [,3] [,4]
wt_healthy   "S"  "I"  "A"  "L"
mutant_tumor "V"  "E"  "R"  "Y"
```

Figure 1: Mutant Cyclin-D1 Model